

SANDESH GHIMIRE

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Google Scholar, Semantic Scholar, DBLP, ResearchGate, LinkedIn

Research Interest

Generative Models, Machine Learning, Deep Learning, Computer Vision, Medical Imaging, Robustness, Statistical Inference, Probabilistic Methods, Quantization and On-device efficiency improvement

Education

PhD in Computing and Information Sciences

Aug 2015 - Aug 2020

Rochester Institute of Technology, NY

GPA: 3.9/4

Advisor: Prof. Linwei Wang, Lab: Computational Biomedicine Lab

B.E. in Electronics and Communication Engineering

Nov 2008 - Oct 2012

Institute of Engineering, Tribhuvan University, Kathmandu, Nepal.

Gold Medalist (Ranked 1/346)

Specialization: Image Processing and Pattern Recognition

Work Experience

Staff Researcher and Engineer

Nov 2025 - Present

Camera Algorithms, Qualcomm Tech Inc.

Senior Researcher and Engineer

Feb 2023 - Nov 2025

Camera Algorithms, Qualcomm Tech Inc.

Supervisor: Hau Hwang

- Leading the development of Small Language Models(SLM) and Small Vision Language Models(SVLM) towards AI on the edge. This includes training LLMs, tokenizers, RL training, evaluations, research, quantization and helping in deployment. PTQ and Quantization Aware Training.
- Working on projects related to Diffusion/Flow Matching and collaborating with university on applied diffusion.
- Collaborating across research teams on research ideas on Gaussian splats based 3D models.
- Investigating how to design computationally efficient models/algorithms without sacrificing their strengths. This includes algorithm improvement and model development: novel architecture design, quantization, distillation.

Postdoctoral Research Associate

Sept 2020 - Feb 2023

Department of Electrical and Computer Engineering, Northeastern University

Supervisor: Profs. Octavia Camps, Jennifer Dy, Dana H. Brooks

- Worked on ML projects related to statistical estimation, interpretability and kernel methods, computer vision projects on video prediction and understanding, diffusion models and energy based models and medical imaging projects on self supervised learning approaches.

Graduate Research Assistant

Aug 2015 - Aug 2020

Computational Biomedicine Lab, Rochester Institute of Technology

Supervisor: Prof. Linwei Wang

- Worked on 1. machine learning and statistical inference with applications to medical imaging, 2. improving generalization of ML models in medical data with semi and unsupervised approaches, 3. geometric deep learning

Research Intern

Jul 2019 - Sept 2019

Medical Sieve Radiology Group, IBM Research, San Jose

Supervisor: Dr. Mehdi Moradi

- On improving generalization of convolutional neural networks while classifying medical images

Electronics/Electrical Engineer

Mar 2014 - June 2015

Nepal Electricity Authority, Central Office, Kathmandu

Research Themes

Computer Vision, Video Analysis and Dynamical System

Northeastern U, Qualcomm

- At Northeastern, I worked on image and video understanding from dynamical system perspective, including latent state space model for video prediction (Koopman perspective). At Qualcomm, I work in generative AI.

Generative Models Vision and Language

RIT, Northeastern U, Qualcomm

- At RIT, I worked on generative models like VAE and GANs in data like spatio-temporal cardiac signals and skin images. At Northeastern and Qualcomm I worked in Diffusion models, theory and its applications for image manipulation tasks. At Qualcomm, I am currently also working on making LLM, VLM and multimodal models compute efficient.

Theoretical Machine Learning

RIT, Northeastern U

- Worked on theoretical machine learning projects like learning theory, generalization bounds, kernel methods, RKHS, measure theoretic approaches, sampling complexity, statistical estimation and interpretability.

Medical Imaging and Biomedical Signal Processing

IBM, Northeastern U, RIT, MSKCC

- Worked on different image analysis and biomedical problems using deep learning/ machine learning on cardiac signals, skin datasets, X-ray images.

Probabilistic Graphical Models and Inference

RIT

- I started my PhD in RIT with PGM framework and Bayesian inference. I modeled the inverse problem of electrophysiological imaging as an inference problem and proposed inference strategies.

Miscellaneous

RIT, Northeastern U

- Worked on semi supervised learning, geometric deep learning, uncertainty estimation, and interpretability of deep learning models.

Research Publications

- ICLR 2026 Regmi, P., **Ghimire, S.** and Li, R., 2026. *Shortcut Diffusion Training with Cumulative Consistency Loss: An Optimal Control View*. ICLR
- CVPR 2024 Liu, J., Teshome, W., **Ghimire, S.**, Sznaier, M. and Camps, O., 2024. *Solving Masked Jigsaw Puzzles with Diffusion Transformers*. CVPR
- AISTATS 2024 Hill, D., Masoomi, A., **Ghimire, S.**, Torop, M. and Dy, J., 2024. *Boundary-Aware Uncertainty for Feature Attribution Explainers* International Conference on Artificial Intelligence and Statistics
- MICCAI 2023 Kumar, N., Gyawali, P.K., **Ghimire, S.** and Wang, L., 2023. *Learning Transferable Object-Centric Diffeomorphic Transformations for Data Augmentation in Medical Image Segmentation* International Conference on Medical Image Computing and Computer-Assisted Intervention
- ICML 2023 Comas, A., Du, Y., Fernandez, C., **Ghimire, S.**, Sznaier, M., Tenenbaum, J.B. and Camps, O., 2023. *Inferring Relational Potentials in Interacting Systems*. International Conference in Machine Learning (**Oral Presentation!**)
- JMI Applegate, M.B., Kose, K., **Ghimire, S.**, Rajadhyaksha, M. and Dy, J., 2023. *Self-supervised denoising of Nyquist-sampled volumetric images via deep learning*. Journal of Medical Imaging, 10(2), p.024005.
- L4DC 2023 Comas, A., **Ghimire, S.**, Fernandez, C., Li, H., Sznaier, M. and Camps, O., 2023. *Learning Object-Centric Dynamic Modes from Video and Emerging Properties*
- NeurIPS 2021 **Ghimire, S.**, Masoomi, A. and Dy, J., 2021. *Reliable Estimation of Kullback-Leibler Divergence using Discriminator in the Reproducing Kernel Hilbert Space*. (**Spotlight talk! - 3% acceptance rate**) [Link to the talk](#)

NeurIPS Workshop 2021	Torop, M., Ghimire, S. , Liu, W., Brooks, D.H., Camps, O., Rajadhyaksha, M., Dy, J. , and Kose, K., 2021. <i>Unsupervised Approaches for Out-Of-Distribution Dermoscopic Lesion Detection.</i>
ArXiv	Ghimire, S., Liu, J., Comas, A., Hill, D., Masoomi, A., Camps, O. and Dy, J., 2023. <i>Geometry of Score Based Generative Model</i>
SPIE	Torop, M., Liu, W., Brooks, D.H., Camps, O., Rajadhyaksha, M., Dy, J. , Kose, K., and Ghimire, S. 2021. <i>Unsupervised representation learning for detecting out of distribution samples in dermoscopy images of eight types of skin lesions.</i>
ICDM 2020	Gyawali, P.K., Ghimire, S. and Wang, L., 2020. <i>Enhancing Mixup-based Semi-Supervised Learning with Explicit Lipschitz Regularization</i>
MICCAI 2020	Jiang, X., Ghimire, S. , Dhamala, J., Li, Z., Gyawali, P.K. and Wang, L., 2019, October. <i>Learning Geometry-Dependent and Physics-Based Inverse Image Reconstruction.</i>
MICCAI 2020	Gyawali, P.K., Ghimire, S. , Bajracharya, P., and Wang, L., 2019, October. <i>Semi-supervised Medical Image Classification with Global Latent Mixing.</i>
TMI 2019	Ghimire, S. , Sapp, J.L., Horáček, B.M. and Wang, L., 2019. <i>Noninvasive Reconstruction of Transmural Transmembrane Potential With Simultaneous Estimation of Prior Model Error.</i>
IPMI 2019	Ghimire, S. , Gyawali, P.K., Dhamala, J., Sapp, J.L., Horacek, M. and Wang, L., 2019, June. <i>Improving generalization of deep networks for inverse reconstruction of image sequences. (Oral presentation - 10% acceptance) IPMI Scholarship Award!</i>
MICCAI Workshop	Ghimire, S. , Kashyap, S., Wu, J. T., Karargyris, A., Moradi, M.,2020. <i>Learning Invariant Feature Representation to Improve Generalization across Chest X-ray Datasets</i>
MICCAI 2019	Gyawali, P.K., Li, Z., Ghimire, S. and Wang, L., 2019. <i>Semi-supervised Learning by Disentangling and Self-ensembling over Stochastic Latent Space.</i>
MICCAI 2019	Dhamala, J., Ghimire, S. , Sapp, J.L., Horáček, B.M. and Wang, L., 2019, October. <i>Bayesian Optimization on Large Graphs via a Graph Convolutional Generative Model: Application in Cardiac Model Personalization.</i>
ICDM 2019	Gyawali, P.K., Li, Z., Knight, C., Ghimire, S. , Horacek, B.M., Sapp, J. and Wang, L., 2019. <i>Improving Disentangled Representation Learning with the Beta Bernoulli Process. (Oral presentation)</i>
CinC 2018	Ghimire, S. and Wang, L., 2018, September. <i>Deep Generative Model and Analysis of Cardiac Transmembrane Potential (Oral Presentation)</i>
MICCAI 2018	Dhamala, J., Ghimire, S. , Sapp, J.L., Horáček, B.M. and Wang, L., 2018, October. <i>High Dimensional Bayesian Optimization of Personalized Cardiac Model Parameters via an Embedded Generative Model. (Oral presentation - 4% acceptance)</i>
MICCAI 2018	Ghimire, S. , Dhamala, J., Gyawali, P.K., Sapp, J.L., Horacek, M. and Wang, L., 2018, September. <i>Generative modeling and inverse imaging of cardiac transmembrane potential.</i>
EP Europace 2018	Cluitmans, M.J.M., Ghimire, S. , Dhamala, J., Coll-Font, J., Tate, J.D., Giffard-Roisin, S., Svehlikova, J., Doessel, O., Guillem, M.S., Brooks, D.H. and Macleod, R.S., 2018. <i>P1125: Noninvasive localization of premature ventricular complexes: a research community based approach.</i>
NeurIPS Workshop	Ghimire, S. , Dhamala, J., Gyawali, P.K., Sapp, J.L., Horacek, M. and Wang, L., 2018, September. <i>Generative modeling and inverse imaging of cardiac transmembrane potential. (Spotlight talk - 6% acceptance)</i>
MICCAI 2017	Ghimire, S. , Sapp, J.L., Horacek, M. and Wang, L., 2017, September. <i>A variational approach to sparse model error estimation in cardiac electrophysiological imaging.</i>
CinC 2017	Ghimire, S. and Wang, L., 2017, September. <i>L0 norm based sparse regularization for non-invasive infarct detection using ECG signal. (Oral Presentation)</i>

CinC 2017 **Ghimire, S.,...**, Wang, L., 2017, September. *Overcoming barriers to quantification and comparison of electrocardiographic imaging methods: A community-based approach*. (Oral Presentation)

Presentations

2023 Image Editing and Enhancement via Image to Image Schrodinger Bridge.
[Talk] Qualcomm AI Summit

2021 Reliable Estimation of Kullback-Leibler Divergence using Discriminator in the Reproducing Kernel Hilbert Space.
[Talk] Neural Information Processing Systems 2021. [Link](#)

2021 Mathematics for Machine Learning
[Talk] Third Nepal Winter School in AI. [Link](#)

2021 On Generalization and Smoothness in Deep Learning
[Talk] Spiral Seminar Series. Northeastern University.

2019 Improving generalization of deep networks for inverse reconstruction of image sequences
[Talk] The 26th International Conference on Information Processing in Medical Imaging (IPMI). (50th Anniversary) 2019. Hong Kong.
[Talk] IBM Research, Almaden, San Jose. 2019.
[Poster] AI@GCCIS Symposium. 2018. RIT
[Poster] Computing Weekend Research Showcase, 2018, RIT

2018 Generative Modeling and Inverse Imaging of Cardiac Transmembrane Potential
[Poster] The 21st International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2018. Granada, Spain.
[Spotlight Talk] NeurIPS Workshop on Machine Learning for Health (ML4H), 2018

2018 Deep Generative Model and Analysis of Cardiac Transmembrane Potentials
[Talk] The 45th Computing in Cardiology Conference. 2018. Maastricht, Netherlands.

2017 A Variational Approach to Sparse Model Error Estimation in Cardiac Electrophysiological Imaging
[Poster] The 20th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2017. Quebec City, Quebec, Canada.
[Poster] GCCIS Research Showcase 2017. RIT
[Poster] Graduate Research Showcase 2017. RIT

2017 Overcoming Barriers to Quantification and Comparison of Electrocardiographic Imaging Methods: A Community-based Approach.
[Talk] The 44th Computing in Cardiology Conference. 2017. Rennes, France.

2017 L0 norm based sparse regularization for non-invasive infarct detection using ECG signal
[Talk] The 44th Computing in Cardiology Conference. 2017. Rennes, France.

Mentoring Experience

Armand Comas With Octavia's supervision, I worked closely with Armand on video interpretability, especially introducing Koopman perspectives to interpret and manipulate dynamic components of a video (see [this paper](#)), and Energy-based models (see [here](#)).

Jinyang Liu I worked closely working with Jinyang on medical image remosaicing and Diffusion-based generative models (see [this paper](#)) under the supervision of Octavia.

Max Torop With Jennifer's supervision, I mentored and worked closely with Max on self supervised learning and out-of-distribution detection topics with the application to medical image analysis. See [this paper](#).

Aria Masoomi I worked closely working with Aria on generalization, kernel methods and interpretable methods under the supervision of Jennifer.

Mentoring Experience

Davin Hill	I mentored and closely worked with Davin on uncertainty and interpretability of deep learning models with the supervision of Jennifer. See this paper
Morgan Kohler	With Octavia's supervision, I worked closely with Morgan mentoring him on the project related to disentanglement and interpretability of video.
Xiajun Jiang	With Linwei's supervision, I mentored and worked with Xiajun on applying graph neural networks to solve inverse problems and electrophysiological imaging. See this paper
Pradeep Bajracharya	With Linwei's supervision, I mentored and worked with Pradeep on Bayesian neural networks and uncertainty quantification.
Nilesh Kumar	With Linwei's supervision, I mentored and worked with Nilesh on learning transformations from data and applying them for data augmentation. See this paper

Awards and Achievements

IPMI Scholarship for junior scientists - awarded to 10 authors at IPMI 2019 conference.

GCCIS Travel Grant by Golisano College of Computing and Information Sciences to present research work at MICCAI 2017

RIT PhD Scholarship/Assistantship 2015-2020. Financial Support for PhD study at RIT.

Prof. F.N. Trofimenkoff Academic Achievement Award for graduating at the top of class (1/346) in B.E. Electronics and Communication, Institute of Engineering.

The College Fellowship (2008-2012) by Institute of Engineering, Central Campus, Pulchowk based on academic merit and performance.

Undergraduate Scholarship (2008-2012) by Institute of Engineering to support tuition during the undergraduate studies at Central Campus, Pulchowk. Rank: 10/12000 in the entrance exam.

Golden Jubilee Scholarship Award 2008 by Government of India based on academic excellence

Mahatma Gandhi Scholarship Award 2006 by Government of India based on academic excellence

Professional Services

Reviewer

Conference: CVPR 2026, CVPR 2024, ICLR 2023, ICML 2022, CVPR 2022, ICLR 2021, Neurips 2021, ICML 2021, AISTATS 2020, MICCAI {2022, 2021, 2020, 2019, 2018, 2017}, MIDL 2020, Women in Machine Learning 2018

Journal: IEEE Transactions on Pattern Analysis and Machine Intelligence, Medical Image Analysis, Medical Physics Journal, Frontiers in Physiology

Associate Editor of Journal, Feb-Dec 2022

Signal, Image and Video Processing. [Link](#)

Organizer / Workgroup Lead

Organization program committee member of the Third Winter School in AI in Kathmandu. [Link](#)

Lead of the workgroup for CEI Pacing Site Localization Challenge 2017 - 2020. [Link](#)

Technical Strengths

Language/Platforms: Python, PyTorch, MATLAB, C/C++

Libraries/Packages: HuggingFace, Scikit-learn, SciPy, Pandas, NumPy, GPyTorch, Matlab-CVX

Tools: L^AT_EX, ParaView, Docker, Git, Conda, Bash